

Antia Lamas-Linares

Profile

Experimental quantum information scientist with a track record of defining and driving product roadmaps across business units and teams. In my most recent role, I have effectively advocated and led the integration of quantum technologies into the AWS networking infrastructure.

My technical expertise is centered around implementations of quantum communication protocols. I demonstrated early efficient free-space entanglement distribution setups and currently develop tailored QKD systems in an industry context. I have worked extensively in security vulnerabilities of real world quantum systems, making contributions that drove change in commercial products and academic implementations. In satellite-based quantum communications, I was Chief Quantum Scientist of Spectral, the only company outside of China that has demonstrated an entanglement source in space, and I am currently supervising the design and build-up of a quantum-enabled optical ground station at the Vigo Quantum Communications Center (Spain).

As a technical lead, I pursue a deliberate strategy to work across organizations by engaging deeply in technical matters and investing in targeted knowledge-sharing practices. This approach has consistently led to joint patent filings and production of technical artifacts with other experts across the company, building trust between the long established engineering organizations and the quantum teams transitioning from a research background. My team gained a reputation for one of the best technical units to interact with, both from other engineering groups, as well as business development, legal, and finance teams. I thrive as a deep technical expert that can spot and pursue connections between loosely related technologies and teams.

Education

1999–2003 **PhD in Physics**, *University of Oxford, UK.*

1998–1999 **MSc in Applied Optics**, *Imperial College of Science and Technology, UK.*

1994–1998 **Licenciatura (BSc) in Physics**, *Universidade de Santiago, Spain.*

Technical Areas

- Design, construction and deployment of complete quantum communications systems in lab and field conditions
- Side channel attacks in QKD systems - quantum hacking

- Quantum time transfer
- Optimization of cryogenic single photon detectors
- Design and construction of entangled photon pair and multi-pair sources

Scientific Management

- Demonstrated success at building effective technical teams and setting up experimental facilities (UK, US and Singapore)
- Trusted C-level engagement partner in deep tech strategy sessions.
- Demonstrated success at attracting academic, startup and industry funding.
- Large international collaboration network across the quantum information field

Experience

- 2021–2026 **Principal Applied Quantum Scientist and Quantum Networking Lead, Amazon Web Services, USA.**
- Initiated activities in quantum networking at AWS
 - Established and led the AWS Center for Quantum Networking
 - Filed seven patents (four granted to date) in QKD, secure randomness generation, hybrid terrestrial and space quantum communications, quantum clock synchronization, entanglement generation and others
 - Oversaw collaborations and POCs with external parties (HSBC, Fidelity, Singapore NQSN) with significant industry and media impact
 - Successfully transitioned QKD from research-oriented quantum organization into network infrastructure organization within AWS
- 2023–present **Visiting Professor, Vigo Quantum Communications Center, Spain.**
Lead of the Satellite Quantum Communications Group. Setting up a quantum-enabled optical ground station for quantum communications applications.
- 2025–present **Advisory board member, Integrated Quantum Networks Hub, UK.**
Advisory board member for the UK IQN hub, the main funding mechanism for UK funding for research in quantum networking.
- 2019–2020 **Chief Quantum Scientist, SpeQtral, Singapore.**
Provide scientific strategic input for Speqtral, a startup focused on delivering quantum communications technology in space.
- 2015–2019 **HPC Researcher, TACC, Austin.**
Member of the High Performance Computing group at TACC, the largest open science computing resource in the USA and home to one of the top 5 clusters in the world.
- Lead on TACCs quantum computing efforts
 - HPC scheduler optimization for heterogeneous systems
- 2010–2020 **Visiting Associate Research Professor, NUS, Singapore.**
Collaborator and consultant for my previous group in Singapore.
- 2010–2012 **Senior NIST-ARRA Fellow, NIST, Boulder.**
Member of the group of Sae Woo Nam, which specializes in extremely sensitive optical cryogenic detectors for quantum information applications.
- 2007–2010 **Principal Investigator, Centre for Quantum Technologies, Singapore.**
Founding PI of CQT and member of the senior team that secured over \$100M in funding.

- 2003–2009 **Assistant Professor**, *National University of Singapore*, Singapore.
Junior faculty member in charge of initiating experimental quantum optics research in NUS.
- 2002–2003 **Postdoctoral Researcher**, *University of California Santa Barbara*, Santa Barbara.
Senior postdoc in charge of facilitating the transition from a UK to a US-based group.

Selected Publications and Patents

Author of 25+ peer reviewed papers with over 4000 citations. Author of 9 patents in quantum information technologies (7 in the last 4 years, 2 of them as an individual inventor, 5 as first author). Full details in google scholar: <https://tinyurl.com/alamaslinares>.

Patents

- Generation of verifiable private randomness using distribution of quantum entanglement, Antia Lamas-Linares, US Patent App. 18542464 (publication - 2025)
- Quantum Safe Enclaves, Antia Lamas-Linares, Pavel Lougovski, Eric Brandwine, US Patent 12,615,140 (granted - 2026)
- Metered Distribution of Quantum Entanglement, Antia Lamas-Linares, US Patent 12,489,612 (granted - 2025)
- Quantum Entanglement Distribution Service, Mihir Bhaskar, et al., US Patent 12,443,874 (granted - 2025)
- Hybrid space-fiber quantum networks for widespread entanglement distribution, Mihir Bhaskar, et al., US Patent 11,962,354 (granted - 2024)
- Quantum secure network clock synchronization, Antia Lamas-Linares and Cyrus Proctor, US Patent 12143203 (granted - 2024)
- Quantum secure clock synchronization based on time-energy and polarization entangled photon pairs, Antia Lamas-Linares and James Troupe, US Patent 11563573 (granted - 2023)

Publications

- A universal setup for active control of a single-photon detector, Qin Liu et al, Review of Scientific Instruments 85, 013108 (2014)
- High quantum-efficiency photon-number-resolving detector for photonic on-chip information processing, B. Calkins et al, Optics Express 21, 22657 (2013)
- Experimentally faking the violation of Bell's inequalities, I. Gerhardt, Q. Liu, A. Lamas-Linares, J. Skaar, V. Scarani, V. Makarov and C. Kurtsiefer, Physical Review Letters 107, 170404 (2011)
- Full-field implementation of a perfect eavesdropper on a quantum cryptography system, I. Gerhardt, Q. Liu, A. Lamas-Linares, J. Skaar, C. Kurtsiefer and V. Makarov, Nature Communications 2, 349 (2011)
- Daylight operation of a free space, entanglement-based quantum key distribution system, M. Peloso, I. Gerhardt, C. Ho, A. Lamas-Linares, C. Kurtsiefer, New Journal of Physics 11, 045007 (2009)
- Testing quantum correlations versus single-particle properties within Leggett's model and beyond, C. Branciard, N. Brunner, N. Gisin, C. Kurtsiefer, A. Lamas-Linares, A. Ling, V. Scarani, Nature Physics 4, 681 (2008)

- Experimental quantum cloning of single photons, A. Lamas-Linares, C. Simon, J.C. Howell, D Bouwmeester, Science 296, 5568 (2002)
- Stimulated emission of polarization-entangled photons, A. Lamas-Linares, J.C. Howell, D Bouwmeester, Nature 412, 6850 (2001)

Other

Languages

English (fluent), Spanish (native), Portuguese (advanced), German (basic)